

Vertical Integration in Uncompetitive Markets: Evidence from the Acquisition of North Star Air by the North West Company

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Abstract

Canada's remote northern communities have some of the highest food prices and food insecurity rates in the world and rely on a small number of retail and transportation firms. In 2017, the largest retailer in the region, the North West Company, acquired a regional airline and started transporting its own air freight to some of the "fly-in" communities it serves. Using a difference-in-difference design, we evaluate the effects of vertical integration on prices and shipment quantities of perishable foods. We find that after the acquisition, communities with vertically integrated retail-air freight experienced 3.8% lower prices and 11.9% higher shipment quantities relative to comparison communities. Effects are qualitatively similar in communities where the acquisition changed firm boundaries without changing routes or upstream competition (Ontario) versus communities where it created new routes and competition (Manitoba and Nunavut). In communities where the retailer has a larger market share, its price falls by less but total community-level shipments rise by more. A theoretical model suggests that this could be due to greater "business-stealing" and lower share of sales exposed to marginal cost reductions in more competitive communities, and evidence from retailer-level shipments supports this interpretation. Our findings show that vertical integration can benefit consumers in markets with limited competition but that the size and nature of these benefits is sensitive to upstream and downstream market structure.

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1 Introduction

Distribution plays a critical role in retail, with the journey from manufacturing plant to store accounting for a large share of value added. Large retailers have increasingly absorbed a larger share of warehousing and distribution in order to improve logistics, reduce costs (through synergies, scale or reduction of double marginalization), and gain competitive advantage against rivals. These aspects of vertical integration have been analyzed theoretically in a general setting (Spengler, 1950; Salinger, 1988; Loudermilk et al., 2023) and evaluated empirically in the context of television (Crawford et al., 2018) and gasoline (Hastings and Gilbert, 2005). In retail and distribution, Amazon represents an extreme case where the retail store is eliminated entirely, but many large retailers have regional distribution centers and their own trucks (Li, 2021). In remote communities, where distribution accounts for an even larger share of value added and there is limited competition in both the freight and the retail sector, many of these rationales for vertical integration may be amplified. This could increase the potential welfare gains from vertical integration, whether it benefits consumers depends on the extent to which marginal cost reductions are passed-through to consumers (Hong and Li, 2017) and the extent to which it promotes anti-competitive practices (e.g. actual or potential retail competitors may be excluded from distribution or receive worse pricing or service quality).

In this paper, we analyze the effects of a vertical integration episode that affected dozens of remote northern communities in Canada. The North West Company (NWC), the dominant retailer in Northern Canada and a retail monopolist in many communities, acquired North Star Air (NSA) in 2017. Before this, NWC contracted with external freight carriers (including NSA) and NSA was only providing passenger and cargo service to communities in Ontario. Subsequent to acquisition by NWC, NSA constructed a new base in Thompson, Manitoba and expanded its freight routes into Manitoba and parts of Nunavut (The North West Company, 2017, 2018; North Star Air, n.d.; Nunatsiaq News, 2018). This acquisition and subsequent developments provide a rich setting to analyze the effects of vertical integration across different market structures. In Ontario markets there was no change in routes or competition in the freight sector but the external contracts between NWC and NSA became internalized, while in Manitoba and Nunavut NSA started new routes that potentially increased competition in the air freight sector. Affected communities in Ontario and Manitoba are mostly NWC retail monopolies, but most Nunavut communities have a major retail competitor (Arctic Co-Operatives Ltd.).

Our analysis focuses on the effects of vertical integration on prices and shipments of perishable foods. Remote “fly-in” communities rely on expensive air freight for these products, making them extremely expensive given their low value to weight ratio. The federal Nutrition North Canada (NNC) program subsidizes shipments of nutritious perishable foods by air and provides data we can use to analyze the effects of vertical integration in this setting. We combine quarterly data on the cost of a food basket collected by NNC (the Revised Northern Food Basket) with monthly data on the weight of subsidized shipments, covering communities currently served by NSA and other remote communities that serve as a comparison group. Using a difference-in-differences design, our main pooled estimates indicate approximately 3.8% lower food-basket prices and 11.9% higher subsidized shipments in NSA-route communities relative to comparison communities following the acquisition and expansion. We distinguish original routes served by NSA in Ontario before the acquisition from the expansion routes into Manitoba and Nunavut. We find that price decreases and quantity

increases for both groups, with somewhat larger effects for the original routes. Empirically our findings are consistent with both reduced double-marginalization and synergies/investments that lower total marginal cost and we cannot isolate a specific mechanism. However, the effects we estimate are not immediate and take a few years after the acquisition to be realized, which suggests that the effect may be more driven by the latter.

We also assess whether price and quantity effects are different across different downstream market structures, i.e. where NWC faces significant retail competition versus where it operates more like a monopoly. In communities where it faces less competition, we find smaller price effects for NWC but larger increases in total community food shipments. We interpret this through the lens of a simple model in which NWC market dominance in a community lowers incentives to pass-through cost reductions but increases the share of retail exposed to a cost reduction, leading to ambiguous effects on quantity. In the expansion communities where NWC faces the most competition, we observe a substantial “business-stealing” effect in that shipments by competitors decrease, which supports this interpretation and indicates that retail competitors do not appear to benefit from NSA’s entry into new freight routes.

We contribute to the empirical industrial organization literature on vertical integration (Hastings and Gilbert, 2005; Hortaçsu and Syverson, 2007; Lafontaine and Slade, 2007; Crawford et al., 2018) by providing the first empirical analysis of vertical integration between a retailer and airline. Our unique setting of remote communities with limited competition allows for a transparent assessment of differential impacts across varying upstream and downstream market structures. Our findings are also of intrinsic interest due to the critical importance of freight costs and food prices in Canada’s remote “fly-in” communities. These communities feature the highest rates of food insecurity in the country and food prices that are often twice as high as the rest of the country in spite of Federal subsidies. The literature finds mixed evidence on whether retail market structure affects subsidy pass-through in this setting (Galloway and Li, 2023, 2026), but subsidy pass-through is regulated, unlike pass-through of other marginal cost reductions. Li et al. (2026) show that freight costs and retailer margins have contributed to higher prices in northern Canada in recent years, but it is not clear what, if anything, can help reduce these. In this context, understanding the impact of market structure and competition is critical for competition policy. Since the North West Company acquisition of North Star Air, there have been two additional mergers of freight carriers in Northern Canada reducing the total number of competing freight carriers in the Ontario, Manitoba and Nunavut fly-in communities to two. These mergers and acquisitions were approved by Canada’s Competition Bureau but to our knowledge there has been no rigorous ex-post evaluation of their effects, making our findings relevant for understanding the competitive implications of future consolidation or entry in the region.

The paper proceeds as follows: we describe the institutional setting, present the conceptual framework, and discuss the data and empirical design. We then report the main results, heterogeneity and retailer-level evidence, and robustness checks. Formal derivations and additional estimates appear in the appendices.

2 Background

2.1 Retail and distribution in Northern Canada

Many communities in Northern Canada are considered “fly-in” communities, meaning that they do not have an all-weather road and rely on a combination of seasonal surface transportation and air freight for goods. Seasonal transportation includes marine transport during the open-water season and ice or winter roads during the winter months. Given the small scale of these communities, perishable food products are often flown in on small aircraft that carry a combination of passengers and cargo. The relatively low value-to-weight ratio of many food products, combined with expensive air freight, contributes to high food prices in communities facing substantial food insecurity. The Canadian government intervenes through its Nutrition North Canada (NNC) program, which subsidizes eligible products based on the weight shipped, with rates that vary by product and community (Indigenous and Northern Affairs Canada, 2017).

Most of these fly-in communities have one or two retail stores, although a handful have three or four. The dominant firm in this setting is the North West Company (NWC), a multinational, publicly listed company whose modern operations originated in the 1987 purchase of the Hudson’s Bay Company’s Northern Stores Division. Many of its northern stores coincide with the locations of historical trading posts. NWC also operates retail businesses in Alaska, the Caribbean, and the Pacific. At January 31, 2018, shortly after the acquisition studied here, its Canadian network included 119 Northern stores and six NorthMart stores, including stores in road-accessible communities (The North West Company, 2018). NWC has a local retail monopoly in several study communities, particularly in Northern Ontario and Manitoba. In Nunavut, it faces significant competition from Arctic Co-operatives Limited, a federation of independent, community-owned co-operatives (Arctic Co-operatives Limited, n.d.). Many communities in the Northwest Territories, Northern Quebec, and Labrador also have an NWC store, with strong competition from a separate co-operative federation in Northern Quebec. Independent stores and small chains provide additional retail alternatives in some communities.

Prior to the acquisition, North Star Air (NSA) was a regional carrier headquartered in Thunder Bay, Ontario, providing passenger and cargo service in Northern Ontario, including the contract for NWC business in the region. Its operations included bases in Pickle Lake, Red Lake, and Sioux Lookout (The North West Company, 2017; North Star Air, n.d.). Regional alternatives included Perimeter Aviation, operating from Winnipeg into Manitoba and Ontario, and Calm Air, serving Manitoba and the Kivalliq region of Nunavut. Perimeter and Calm Air belonged to the same parent company, Exchange Income Corporation (EIC). NWC stores in Manitoba and the neighbouring Kivalliq region of Nunavut were served by Calm Air on dedicated freighters prior to the acquisition. In Nunavut and the Northwest Territories, First Air and Canadian North were important passenger and freight providers before their 2019 merger (Competition Bureau, 2017, 2019).

While we can observe historic passenger routes of these airlines, scheduled passenger service is an imperfect measure of the freight alternatives available to retailers. Limited overlap between passenger networks does not necessarily imply that a retailer has access to only one freight provider: charter flights and dedicated freight contracts can provide alternatives that are not apparent from scheduled passenger routes. This distinction matters in our setting because Calm Air’s NWC business in Manitoba and Kivalliq

used dedicated freight charters rather than its combination passenger-and-freight aircraft (Exchange Income Corporation, 2017), but Calm Air and Perimeter continued to operate combination aircraft (and provide a competitive freight alternative) thereafter.

One important development during this period, which may have served as a catalyst for subsequent changes in distribution, was the introduction and expansion of NNC. NNC was introduced in 2011 to replace “Food Mail,” a subsidized shipping service offered through Canada Post. Under Food Mail, the government subsidized Canada Post to provide shipping rates that were generally below the commercial cost of transporting food to eligible communities. Under NNC, registered retailers and suppliers instead receive a specified subsidy per kilogram of eligible products shipped irrespective of carrier (Indigenous and Northern Affairs Canada, 2017; Crown-Indigenous Relations and Northern Affairs Canada, 2020). Eligibility expanded to additional communities, including some communities served by NSA in Manitoba and Northern Ontario, in October 2016. Further subsidy-rate increases followed in January 2019, May 2020, and February 2023 (Indigenous and Northern Affairs Canada, 2016; Galloway and Li, 2026; Nutrition North Canada, 2025). In January–March 2023, NNC-eligible communities in Ontario, Manitoba, and Nunavut together received approximately \$23.8 million in subsidies, accounting for about 70.5% of the national total (Nutrition North Canada, 2025).

Because accessing the Food Mail subsidy required shipping through Canada Post, retailers sometimes made their own freight arrangements where private freight costs were lower than the subsidized postal rate, including in parts of Ontario and Manitoba. Retailers argued that receiving the subsidy directly would allow them to negotiate better freight costs, improve logistics, and lower food prices (Standing Committee, 2011). A government evaluation subsequently reported that interviewed retailers attributed improvements under NNC to their ability to negotiate freight rates directly, while noting that this opportunity remained limited in communities with few suitable carriers (Crown-Indigenous Relations and Northern Affairs Canada, 2020).

There is evidence consistent with these claims: Georgiev and Li (2026) and Crown-Indigenous Relations and Northern Affairs Canada (2020) find that the transition from Food Mail to NNC lowered prices for broad food baskets, despite being approximately budget-neutral from the federal government’s perspective.¹ The transition gave retailers greater incentives to optimize their freight costs since they were no longer constrained to use Canada Post as a freight carrier, and one way of doing this may have been through acquisition of an airline.

2.2 The North West Company acquisition of North Star Air

The agreement for NWC to acquire NSA for approximately \$31 million was announced on April 28, 2017. NWC also announced plans to invest a further \$14 million in additional aircraft and hub capacity (The North West Company, 2017). At the time of the announcement, EIC reported that Calm Air had received 90 days’ notice that NWC would transfer its freight business in Manitoba and Kivalliq to NSA. That business generated approximately \$14 million in annualized revenue for Calm Air (Exchange Income Corporation, 2017).² The

¹ By contrast, Fafard St-Germain et al. (2019) and Daley et al. (2024) find increases in food insecurity for Food Mail eligible communities following the transition, but do not provide evidence for a retail price mechanism.

² EIC described the contract as the most competitively priced/lowest margin contract in their portfolio, such that the anticipated earnings loss of the contract would be less than \$1 million on \$14 million in revenues. It stated that the change would not affect the

acquisition was completed on June 15, 2017, creating an integrated retailer-carrier organization (The North West Company, 2018).

The subsequent expansion involved both new destinations and investment in the freight network. A report published on August 21, 2018 described the opening of NSA's new hangar and warehouse in Thompson, Manitoba, to support service to Northern Manitoba and the Kivalliq region of Nunavut, along with several communities in the Qikiqtaaluk (Baffin) and Kitikmeot regions of Nunavut. The report also described an expansion of NSA's fleet from 11 to 16 aircraft and stated that the airline was serving approximately 50 remote communities (Nunatsiaq News, 2018). NSA's own history likewise links the acquisition to additional aircraft, the Thompson base, and expansion into Manitoba and Nunavut (North Star Air, n.d.).

Figure 1 shows the geographic distribution of the communities included in our maps and their NSA route exposure. It distinguishes 24 original NSA communities in Ontario, 26 expansion NSA communities, and 49 control communities, together with six NSA air bases. The original routes are concentrated in Northern Ontario, while the expansion routes extend into Manitoba and Nunavut (but not the far Western and Eastern parts of Nunavut). Controls are distributed across the North, including both communities near the treated regions and communities farther away. The figure identifies community locations and exposure groups; it does not draw individual flight paths or imply that every community is served directly from its nearest displayed base.

We classify as "original routes" the communities served by NSA before the acquisition. Here, the acquisition changed the ownership of an existing freight and internalized a previously external relationship between the retailer and airline without introducing new competition. The remaining treated communities are classified as "expansion routes," reflecting NSA's post-acquisition network expansion. In these communities, other freight carriers continued to operate but now faced additional competition from NSA, which offers third-party freight service as well as transporting freight for NWC. The "treatment" here combines internalization of freight services with competitive entry and investment in additional capacity. Note that NSA only offers passenger service in some of these communities, and that it offers freight service to some Ontario communities that do not have an NWC store.

The expansion routes exclusively connect communities with an NWC store, as NSA only carries freight in these regions and has limited incentives to run regular freight to communities with no NWC store. For example, there is no record of regular NSA service to Whale Cove (Tikrarjuaq), which has no NWC store, despite being closer than many other NSA communities from the Thompson base. The precise timing of the expansion is less clear than the acquisition date. NWC gave notice of the Calm Air contract termination in spring 2017, while the new Thompson facilities opened in August 2018. The extent to which charters, aircraft operating from Ontario hubs, or interim Thompson operations were used between these dates is not fully documented. We use archived websites and news reports to classify service routes, and these establish that the expansion destinations on the map were served by November 2018.³

efficiency or profitability of its combination-aircraft operations or its separate long-term freight contract with Arctic Co-operatives (Exchange Income Corporation, 2017).

³Our archival notes record third-party cargo service to Manitoba destinations through Red Lake or Thompson by May 2018, alongside a route map listing destinations in Kivalliq and elsewhere in Nunavut. Most of the "treated" Nunavut communities in our map and classification were listed on the November 2018 third-party cargo page and in the May 2018 map, and are still listed on the website today. The exceptions are Coral Harbour and Nauyasat, which first appear in a September 2024 capture. As these two Kivalliq

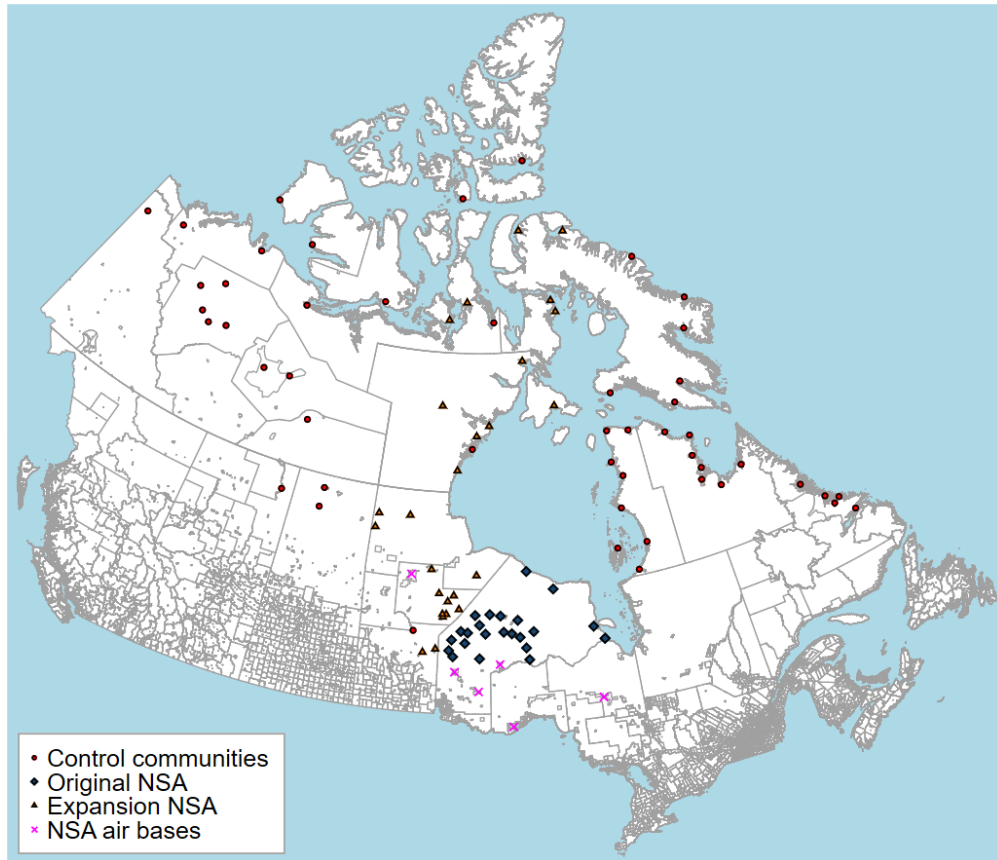


Figure 1: Study communities, NSA route exposure, and air bases

Notes: Red circles denote control communities; navy diamonds denote original NSA communities; orange triangles denote expansion NSA communities. Magenta crosses denote Thunder Bay, Sioux Lookout, Pickle Lake, Red Lake, Kapuskasing, and Thompson air bases. The map displays 99 communities: 49 controls, 24 original NSA, and 26 expansion NSA. The six bases are additional points, not part of these community counts. Background polygons are Statistics Canada's 2021 census divisions.

Figure 2 adds the downstream retail dimension to this geography. It shows the same communities and air bases, but colors each community according to NWC's baseline share of subsidized shipment weight. Shapes retain the route classifications from Figure 1: circles are controls, diamonds are original NSA communities, and triangles are expansion NSA communities. Orange indicates an NWC share below 70%, while forest green indicates a share of at least 70%. For communities with an NWC store, this is its median market share, so these roughly split the sample into an equal number of high and low market share communities.

The figure shows a relationship between route exposure and NWC's initial shipment share and variation within each route group. NWC accounts for at least 70% of baseline shipments in 17 of the 24 original-route communities and 19 of the 26 expansion-route communities, compared with 8 of the 49 controls. Importantly, neither treated group consists entirely of high-share communities. This variation motivates examining whether

destinations were served by the canceled Calm Air contract, are within range of the Thompson base, and have an NWC store, we classify them as treated expansion communities. On the other hand, we exclude one community that is currently listed – Kugaaruk, which did not have an NWC store until November 2023. Resolute (which has no NWC store) and Cambridge Bay (which does) were briefly listed in 2018 but were no longer listed by June 2019 and are not listed today. We classify these as untreated.

price and quantity responses differ with the local retail environment. Note that our shipment data only cover subsidized products, not the total market share for food or retail.

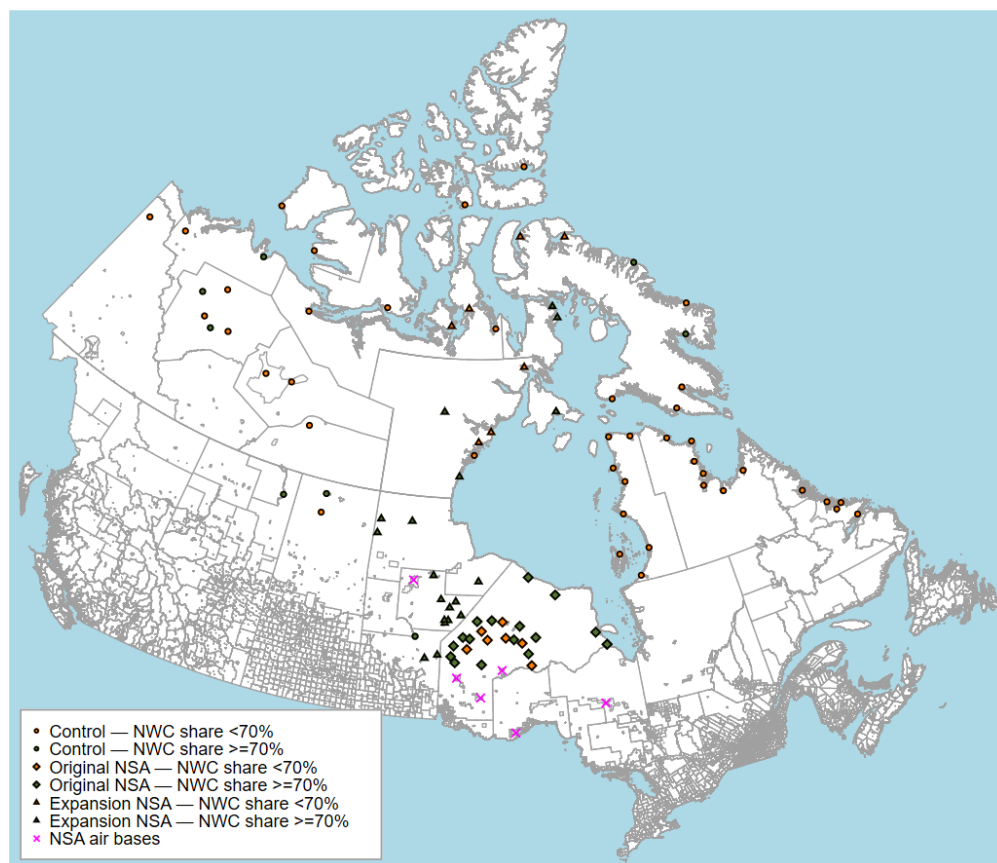


Figure 2: NSA route exposure and baseline NWC shipment share

Notes: Communities and bases are the same as in Figure 1. Circles denote controls, diamonds denote original NSA communities, and triangles denote expansion NSA communities. Orange markers indicate baseline NWC shares of subsidized shipment weight below 70%; forest-green markers indicate shares of at least 70%. Magenta crosses denote NSA air bases, to which the share classification does not apply. Background polygons are Statistics Canada's 2021 census divisions.

Subsequent to the NWC–NSA acquisition, there was further consolidation among competing airlines. In 2019, Canadian North, owned by the Inuvialuit Regional Corporation, and First Air, owned by Makivik Corporation, merged. The Competition Bureau identified substantial overlap and raised concerns that the transaction would eliminate competition in several affected markets (Competition Bureau, 2019). Following a public-interest assessment incorporating the Bureau's findings, the Government of Canada approved the merger subject to conditions covering passenger and cargo prices, service schedules, access to northern infrastructure, and monitoring (Transport Canada, 2019). After the COVID-19 pandemic, the government and Canadian North agreed on revised conditions in April 2023. Transport Canada cited the airline's financial position and its ability to comply with the original requirements (Transport Canada, 2023).

EIC subsequently announced an agreement to acquire Canadian North for \$205 million on February 24,

2025; the transaction closed on July 2, 2025 (Exchange Income Corporation, 2025). EIC, already the owner of Calm Air and Perimeter Aviation, described Canadian North's routes as complementary to its existing northern network, and the merger did not eliminate competition on any routes. However, it transferred ownership from Inuit Treaty corporations to a publicly traded corporation with potentially different incentives and substantial additional aviation assets. These transactions highlight the ongoing, potentially reactive, changes in the northern freight market, but are too recent to study here.

Taken together, our description of the institutional context and our two maps highlight that the acquisition need not have had a uniform effect across communities. Original routes primarily expose an existing network to a change in ownership, whereas expansion routes combine integration with new freight service and network investment. Within those groups, NWC's retail position also varies. These differences motivate the separate analysis of route exposure and downstream retail competition in the sections that follow.

3 Conceptual Framework

We use a retail-pricing model to examine how NWC's acquisition of North Star Air (NSA) and the airline's subsequent expansion could affect food prices and shipments. We represent the possible benefits of these changes as a reduction in NWC's cost of supplying an additional unit of food. We then compare how NWC responds when it is the only local retailer and when it faces one competitor. In each case, we compare outcomes before and after the same cost reduction, with no exit/entry. NSA enters the model through its effect on NWC's costs; we do not model the airline's decisions separately. Appendix A derives the predictions.

NWC could benefit from the acquisition in two ways. First, when the retailer and airline are separate, the freight price paid by NWC includes the airline's profit margin. Bringing them under common ownership allows NWC to consider the actual cost of carrying its goods rather than treating that margin as a payment to another company (Spengler, 1950). Although it is possible that NWC-NSA set marginal freight costs jointly to maximize total surplus through various mechanisms, if NSA set freight prices above its marginal costs this would result in a final price above the profit-maximizing price, lowering total profits. Internalization would remove this distortion. Second, better flight schedules, aircraft use, warehousing, and delivery coordination can reduce the actual cost of supplying stores. On expansion routes, new service and additional capacity may offer NWC a cheaper way to transport food. Our theoretical analysis below takes the size of the resulting cost saving as given and examines how it affects retail decisions, but in principle cost savings could be different in the original versus the expansion routes.

Each retailer sets its price to maximize profit. We assume linear demand: holding other prices fixed, equal changes in a store's price produce equal changes in how much customers buy. When two stores operate, they choose prices at the same time, and customers can switch between them. The stores can differ in location, service, and product selection, so shoppers do not choose solely on price (Singh and Vives, 1984). Sales at each store respond more strongly to its own price than to the other store's price. The retailers can also have different costs and shares of sales. In the basic model, only NWC's cost falls. The competitor's cost and access to freight stay the same, and both stores remain open.

When NWC is the only retailer, it passes half of the cost saving to consumers through a lower price

and keeps the other half as a larger profit margin on each unit sold. The cost on every unit shipped to the community decreases, and all of NWC's additional sales represent an increase in total community purchases.

When NWC faces a competitor, the two stores respond to each other's prices. NWC's lower price encourages its competitor to lower its price, and that response gives NWC a further reason to cut its own price. Under the model's assumptions, the same cost saving therefore produces a larger reduction in NWC's price when it faces competition.

However, only some of the units shipped to the community experience a reduction in costs. Some of NWC's additional sales come from customers switching stores. Although the competitor also lowers its price, NWC's price falls further, so the competitor loses sales. NWC's sales therefore increase by more than total community purchases. This can happen even though the competitor's costs and freight service have not changed, i.e. without foreclosure.

Total purchases increase in both cases, which can be interpreted as evidence for increased consumer welfare. Whether total purchases increase more under monopoly or competition depends on how customers respond to prices. The larger price reduction in competitive markets may increase NWC's sales by more, but the loss of sales at the other store may offset part of that increase. When customers' responses to price changes are the same at both stores, the model predicts a smaller increase in total purchases under competition. When those responses differ, either setting can produce the larger increase. The main point is that a larger reduction in NWC's price is not a sufficient statistic for the consumer welfare impact, and total community purchases provides a better reflection of the equilibrium effect on consumer welfare.

This distinction matters for the empirical analysis because the RNFB price observations primarily reflect NWC prices, while total shipment quantities include other retailers and purchasing channels. We therefore examine NWC's prices, its own shipments, and total community shipments separately in light of the theoretical framework described above. The basic model predicts that NWC gains sales while its competitor loses some. An extension in the appendix allows the competitor's costs to fall as well and shows how both retailers can then expand. This provides another possible explanation for changes in shipments when freight improvements benefit more than one store. The appendix also examines the broader economic effects by considering these consumer gains alongside changes in firms' profits and the costs of supplying food.

4 Data

4.1 Price and quantity data from NNC

The price outcome we analyze is the cost of the Revised Northern Food Basket (RNFB). The RNFB is a standardized, fixed-weight basket of 67 items designed to provide adequate nutrition for a family of four for one week. Nutrition North Canada (NNC) uses the basket to monitor food costs in northern communities, and aggregates prices sent directly by retailers. RNFB costs are publicly reported at the community-by-quarter level for most NNC-eligible communities during our sample period (Nutrition North Canada, n.d.; Indigenous and Northern Affairs Canada, 2017). Approximately 75% of its total product weight consists of perishable goods that are shipped by air and eligible for the NNC subsidy, providing an informative measure of food prices most impacted by air freight costs and subsidies. However, the RNFB is not a representative price

index and excludes many important foods. We use the natural logarithm of the dollar value of the RNFB cost as our main price outcomes.

The store coverage of the price data is an important consideration. Before October 2023, most independent stores and Arctic Co-operatives stores were not included in the RNFB reporting used here. Consequently, for most communities – and all treated communities – the reported prices reflect NWC rather than a comprehensive average across all local retailers.⁴ The absence of comprehensive, disaggregated prices for all retailers and products limits our ability to distinguish NWC’s price response from changes in community-wide prices.

The quantity outcome is constructed from NNC administrative records of subsidized shipments. These records identify shipment activity by community and reporting period and can be disaggregated by broad product category and recipient. They are more comprehensive than the RNFB price data in that they cover recorded subsidized shipments for all retailers and subsidized products, rather than prices from only from select retailers like NWC. But like the RNFB, they exclude many important food products that are not subsidy eligible.⁵ We use the natural logarithm of total subsidized shipment weight or per capita kilograms of shipments (using 2016 census population counts) as the aggregate quantity outcome. In our estimation, we impose minimum shipment restrictions to exclude communities that only make minor use of the subsidy (e.g. communities that only use it during brief seasonal periods when surface transportation is not available, communities that are eligible but the retailer has not registered to receive the subsidy, and the period before October 2016 when some communities were not yet eligible).

We also use NNC administrative data to estimate the subsidy content of the RNFB or the average subsidy level, which serve as important control variables given changes in subsidy rates in October 2016, January 2019, May 2020 and February 2023. Following Galloway and Li (2023), we apply the relevant product-level subsidy rates to the corresponding weights in the basket for each community and period to generate the RNFB subsidy, and we use the total subsidy value divided by shipment weight to estimate the average subsidy per kilogram. Baseline subsidy rates also provide a proxy for relative remoteness and hub-distance in the cross-section; subsidy rates are partly determined at a regional level and are not a direct measure of cost, but are strongly correlated with observable differences in freight costs (Li et al., 2026). This allows us to flexibly control for time-varying factors that might influence communities differently depending on flight length, such as energy prices.

To assess retail market structure, we use the distinction between clients and recipients in the NNC shipment records. The client classification distinguishes retailers from individuals and organizations, while the recipient identifies the firm receiving the subsidy. We identify an NWC retail presence from shipments for which NWC is the recipient. Shipments from another recipient to a retail client provide evidence of competing local retail activity, rather than direct ordering by individuals or organizations.

⁴The exceptions in our sample include Nunavik, Quebec, where the co-operative federation FCNQ is included in the community-average calculation for all 14 communities and, in three cases, represents the only store in the community. In three Northwest Territories communities, the RNFB is an average of NWC and either Stanton or Rampart. In Yukon, the sole NNC-eligible community receives special subsidies and its NWC store was replaced by a co-operative in 2014 (this does not affect our analysis). Before the 2016 fiscal year, another store in Nain, Newfoundland and Labrador, was also included in the reported average.

⁵This includes products that are subsidy-eligible when shipped by air but not eligible when shipped by surface transport, such as flour. Following the introduction of a surface-transport subsidy in January 2019, the data also record surface shipments of eligible products, although these quantities are small in the records examined for this study.

In some cases, the recipient identifies the local store directly, as with a community co-operative. In other cases, a recipient may supply several independent stores, or different recipients may supply the same local retail market. Consequently, the number of subsidy recipients should not be interpreted mechanically as the number of local stores. We distinguish shipments associated with NWC, other local retailers, and direct orders by individuals or organizations through “Southern suppliers.” This classification is important because Southern suppliers can provide an alternative to local retail purchases without necessarily operating a store in the community.

4.2 Sample composition

Table 1 summarizes the dataset we use for our main estimates. Panel A reports community and observation counts separately for prices and quantities. The quantity sample contains 99 communities: 49 controls, 24 original NSA communities, and 26 expansion NSA communities. The price sample contains 89 communities: 45 controls, 18 original NSA communities, and 26 expansion NSA communities. There are 5,219 price observations and 11,727 quantity observations. The original and expansion groups are mutually exclusive, and the “All communities” row combines the three groups.

Both samples require community-average shipment weight per resident and shipment weight per resident in the observation period to exceed 15 kilograms, and exclude the phase-in period.

Panel B reports unweighted observation-level means within these samples. The average weekly RNFB cost is \$494.98. It is highest among controls, at \$501.43, compared with \$491.53 in original NSA communities and \$488.21 in expansion NSA communities. Mean monthly shipment weight per resident is 28.27 kilograms overall: 27.72 in control communities, 30.57 in original NSA communities, and 27.47 in expansion NSA communities. These statistics describe observed levels across the route groups over the included pre- and post-treatment periods.

Table 1: Sample composition and descriptive statistics

<i>Panel A. Sample coverage</i>				
Treatment group	Quantity-sample communities	Price-sample communities	Price observations	Quantity observations
All communities	99	89	5,219	11,727
Control communities	49	45	2,398	5,852
Original NSA	24	18	1,097	2,583
Expansion NSA	26	26	1,724	3,292

<i>Panel B. Outcome means</i>				
Treatment group	RNFB (\$ per week)	Log RNFB	Shipment kg/person	Log total shipment weight
All communities	494.98	6.19	28.27	9.86
Control communities	501.43	6.20	27.72	9.67
Original NSA	491.53	6.19	30.57	9.80
Expansion NSA	488.21	6.18	27.47	10.24

Notes: Counts and means use the actual estimation samples of the main price and quantity regressions, respectively. Both samples require community-average and contemporaneous shipment weight per resident above 15 kilograms and exclude the phase-in period. RNFB is the weekly dollar cost of the Revised Northern Food Basket. Shipment kilograms per person divide total monthly community shipment weight by population. Log RNFB and log total shipment weight are natural logarithms. Original NSA, expansion NSA, and control communities are mutually exclusive groups.

5 Empirical Strategy

5.1 Difference-in-differences

We estimate how prices and subsidized shipments change in NSA-route communities relative to control communities. The main outcomes are the natural logarithm of the RNFB cost and the natural logarithm of total subsidized shipment weight. We also examine RNFB prices in dollars and shipment quantities in kilograms per resident. The dataset is indexed by community and calendar month. Price regressions use observations for which RNFB prices are available, while shipment regressions use the monthly shipment records.

Let O_c indicate an original NSA community and E_c an expansion NSA community. These groups are mutually exclusive. The pooled NSA indicator is $A_c = O_c + E_c$; control communities have $A_c = 0$. These indicators describe community-level route exposure and are distinct from the post-treatment indicator, $Post_t$, which varies over time.

For outcome Y_{ct} in community c and period t , the pooled specification is

$$Y_{ct} = \alpha_c + \delta_t + \beta (A_c \times Post_t) + X'_{ct} \gamma + \varepsilon_{ct}, \quad (5.1)$$

where community fixed effects α_c absorb persistent differences in geography, transportation access, and retail market structure. Period fixed effects δ_t absorb seasonality and other shocks common to communities in a given calendar period. The vector X_{ct} contains the additional controls described below.

To distinguish integration on an existing network from post-acquisition expansion, we replace the pooled

interaction with separate original-route and expansion-route interactions:

$$Y_{ct} = \alpha_c + \delta_t + \beta_O (O_c \times Post_t) + \beta_E (E_c \times Post_t) + X'_{ct}\gamma + \varepsilon_{ct}. \quad (5.2)$$

Both interactions enter the same regression, leaving control communities as the omitted route category. Thus, β_O and β_E measure the respective changes in original and expansion NSA communities relative to controls. The pooled specification summarizes the changes across both treated groups, while the disaggregated specification allows their responses to differ. The post indicator is coded to begin in August 2017, following the notice period for transferring NWC's freight business from Calm Air to NSA.

The benchmark regressions exclude the transition period from August 2017 through August 2018, when the network was being expanded and the Thompson facilities were being developed. Consequently, the retained observations compare periods up to July 2017 with periods from September 2018 onward.

The resulting coefficients describe longer-term changes associated with integration and the subsequent network configuration, rather than an immediate response on the legal acquisition date. We supplement our difference-in-difference analysis with event-study plots that include the transition period and shed light on any potential anticipatory or delayed responses of prices and quantities.

We begin with a simple specification that includes only community and period fixed effects, then add controls for changes in NNC subsidies. Price regressions control for the estimated subsidy content of the RNFB, while shipment regressions control for the community-period subsidy-per-kilogram measure which is a better indicator of the subsidy level of all subsidized shipments. Our most saturated specification also includes the interaction of each community's baseline subsidy rate (we use the level 1 NNC subsidy) with period indicators. These interactions allow the relationship between initial subsidy rates and outcomes to vary across calendar periods, and are meant to capture factors like energy costs that might vary over time and differentially affect freight costs across communities in a way that is correlated with the NSA acquisition. The benchmark sample requires shipment weight per resident to exceed 15 kilograms in the observation period and average shipment weight per resident to exceed 15 kilograms over the available community observations. These thresholds correspond to approximately the 10th percentile of the monthly shipment distribution, and we consider robustness to other thresholds.

Regressions are unweighted, and standard errors are clustered by community. The corresponding tables report observation counts and the controls included in each specification. For logarithmic outcomes, coefficients describe changes in log points and approximate proportional changes for small estimates. The level specifications express changes in dollars per RNFB basket or kilograms shipped per resident. The identifying assumption is that, conditional on the fixed effects and included controls, outcomes in NSA-route communities would have evolved like those in control communities in the absence of the acquisition and associated network changes. Persistent differences in outcome levels do not by themselves violate this assumption; differential changes unrelated to treatment would.

The interpretation also requires attention to spillovers. Changes in freight competition or NWC's wider distribution network could affect communities outside the treated route groups. If control communities are affected, the estimates measure changes relative to those affected controls rather than the full effect relative

to an unaffected counterfactual. The comparison between original and expansion routes does not eliminate this concern. We therefore interpret the estimates alongside the dynamic evidence and the sensitivity of the results to alternative sample and comparison-group restrictions.

5.2 Dynamic effects/event study

We examine the timing of outcome changes using interactions between route-group indicators and calendar-period indicators. For the disaggregated specification, we estimate

$$\begin{aligned}
Y_{ct} = & \alpha_c + \delta_t \\
& + \sum_{g \in \{O, E\}} \sum_{\tau \neq \tau_0} \beta_{g\tau} [G_{gc} \times \mathbf{1}\{t = \tau\}] \\
& + X'_{ct} \gamma + \varepsilon_{ct},
\end{aligned} \tag{5.3}$$

where $G_{Oc} = O_c$, $G_{Ec} = E_c$, and τ_0 is the omitted reference period for the outcome being examined. The pooled specification replaces the separate route-group interactions with interactions using A_c . In the disaggregated specification, the original and expansion interactions enter jointly, with control communities providing the comparison group.

The event study plots treat June 2017, when the acquisition was legalized, as the beginning of the post period. For prices, we use RNFB costs in dollars in each quarter, with March 2017 (the last price reporting period before acquisition) normalized to zero. The shipment event study uses monthly kilograms per resident and normalizes June 2017 to zero.

Each coefficient measures the change in the treated-control outcome difference relative to the relevant reference period. Unlike the benchmark before-and-after regressions, the dynamic specifications retain the transition-period observations. This allows the figures to show whether changes emerge during the network expansion or only after the Thompson facilities opened. The specifications include community and period fixed effects, the outcome-specific subsidy control, and baseline-subsidy-rate-by-period interactions. Confidence intervals are based on standard errors clustered by community.

The dynamic estimates are informative about both pre-acquisition movement and the evolution of post-acquisition differences. Note that individual months of data yield noisy estimates, which is part of the rationale for pooling before and after periods to estimate the difference-in-difference specification, and given the noise, one should not interpret the significance of individual coefficients as evidence for against pre-trends or average differences between the pre and post period (Roth et al., 2023). Systematic pre-trends would make a causal interpretation less plausible.

5.3 Heterogeneity

To examine whether responses vary with NWC's competitive position in local market market, we interact treatment with a variable S_c that captures differences across communities. For the pooled specification,

$$\begin{aligned} Y_{ct} = & \alpha_c + \delta_t + \beta (A_c \times Post_t) \\ & + \theta (S_c \times Post_t) + \phi (A_c \times Post_t \times S_c) \\ & + X'_{ct}\gamma + \varepsilon_{ct}. \end{aligned} \tag{5.4}$$

The time-invariant terms involving A_c , S_c , and their interaction are absorbed by community fixed effects. The $S_c \times Post_t$ term allows communities with different shipment shares to experience different before-and-after changes even outside the treated group.

Our main measure of retail market heterogeneity S_c is an indicator equal to one if NWC's baseline market share (between October 2016 and May 2017) is greater than 0.7 and zero otherwise. The 0.7 threshold is approximately the median NWC sample share in sample communities with an NWC store. For communities with a baseline market share below 0.7, the estimated treatment effect is β , while for those with share above 0.7, the estimated treatment effect is given by $\beta + \phi s$. For a negative price effect, a positive interaction coefficient indicates a smaller price reduction at high NWC market shares. For robustness, we can consider a continuous measure of baseline market share, but using an indicator that splits the sample at 0.7 increases statistical power; as illustrated in our map, it still leaves substantial variation in low/high market share communities across both original and expansion provinces.

Finally, we examine how shipment changes are distributed between NWC and other suppliers. Using the same pooled and joint original/expansion specifications, we replace the aggregate outcome with NWC shipments per capita and competing local-retailer shipments per capita. An additional outcome includes all shipments outside NWC, encompassing both competing local retailers and direct orders. This broader non-NWC measure contains, rather than being separate from, the local-retailer component.

These decompositions distinguish NWC's own expansion from changes in other shipment channels. They also connect the empirical analysis to the conceptual framework: stronger NWC price reductions can increase NWC shipments while diverting some purchases from competitors, leaving a smaller change in aggregate shipments. A decline in rival shipments does not, by itself, identify foreclosure. Without observations on freight contracts, access, prices, and service quality, the analysis cannot determine whether such a decline reflects business stealing, changes in rivals' input terms, or other adjustments.

6 Main Results

6.1 RNFB prices

Table 2 reports the price estimates. Columns (1)–(3) pool all NSA-route communities; columns (4)–(6) jointly distinguish original and expansion routes. The pooled log-price coefficient is negative throughout, falling in magnitude from -0.064 with only community and period fixed effects to -0.039 after adding the RNFB subsidy and baseline-subsidy-rate-by-period controls. The latter estimate corresponds to approximately 3.8%

lower RNFB prices and is statistically significant at the 5% level. The attenuation indicates that the additional controls account for part, but not all, of the initial treated-control difference in price changes.

In the route-specific specification with all controls included, original NSA communities have an estimated price decline of 0.053 log points, approximately 5.2%, while expansion communities have a decline of 0.035 log points, approximately 3.4%. Both estimates are statistically significant. However, the larger original-route point estimate does not establish that the effects differ: the coefficient-equality test in Appendix Table B.4 has a p -value of 0.279.

Panel B provides the corresponding dollar estimates. The fully controlled pooled coefficient is $-\$17.96$ per weekly basket. The original- and expansion-route coefficients are $-\$20.05$ and $-\$17.45$, respectively, significant at the 10% and 5% levels. Thus the negative price association is not specific to the logarithmic transformation. Original routes are the closer comparison to an ownership change on an existing network, whereas expansion-route estimates combine integration with entry and network investment. Neither comparison separately identifies the contribution of internalized freight margins and operating cost reductions.

Table 2: Effects of NSA route exposure on RNFB prices

	All NSA routes			Original vs. expansion		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Log RNFB price</i>						
Post $\times$ NSA	-0.064*** (0.015)	-0.061*** (0.014)	-0.039** (0.015)			
Post $\times$ original NSA				-0.094*** (0.023)	-0.090*** (0.022)	-0.053*** (0.020)
Post $\times$ expansion				-0.051*** (0.015)	-0.049*** (0.015)	-0.035** (0.015)
Observations	5,219	5,219	5,219	5,219	5,219	5,219
Adjusted R^2	0.926	0.927	0.936	0.927	0.928	0.936
<i>Panel B. RNFB price in dollars</i>						
Post $\times$ NSA	-31.790*** (7.372)	-30.886*** (7.402)	-17.957** (7.690)			
Post $\times$ original NSA				-44.083*** (12.002)	-42.988*** (11.967)	-20.052* (10.549)
Post $\times$ expansion				-26.293*** (7.565)	-25.699*** (7.647)	-17.446** (7.895)
Observations	5,219	5,219	5,219	5,219	5,219	5,219
Adjusted R^2	0.918	0.918	0.930	0.919	0.919	0.930
Community fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
RNFB subsidy control	No	Yes	Yes	No	Yes	Yes
Baseline rate $\times$ period	No	No	Yes	No	No	Yes

Notes: Columns (1)–(3) pool all NSA route communities. Columns (4)–(6) include the original- and expansion-route interactions jointly. Our sample restricts to periods with above 15 kilograms of shipments per capita and communities with an average of at 15 kilograms per capita, and excludes the phase-in period. All specifications include community and period fixed effects. Columns (2), (3), (5), and (6) control for the estimated RNFB subsidy; columns (3) and (6) additionally include baseline-rate-by-period interactions. Standard errors clustered by community are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.2 Subsidized shipment quantities

Table 3 reports the aggregate shipment estimates. In Panel A, the pooled log-weight coefficient declines from 0.107 in column (1) to 0.095 in column (3), corresponding to approximately 9.5% higher shipments in the specification with all controls included. The original-route coefficient is 0.124 (approximately 12.4%), while the expansion-route coefficient is 0.083 (approximately 8.3%) in column (6). All three fully controlled estimates are statistically significant at the 1% level.

Panel B shows positive changes in shipment kilograms per resident for both route groups. The estimates with all controls included are 2.859 kilograms per resident for pooled NSA exposure, 4.225 for original routes, and 2.297 for expansion routes, all significant at the 1% level. The logarithmic and per-capita specifications provide complementary measures: one measures proportional changes in total shipment weight, while the other measures absolute changes per resident.

All specifications use the same 11,727 observations from 99 communities. Although the estimated proportional increase is larger on original routes, the coefficient-equality test in Appendix Table B.4 does not reject equality between the original- and expansion-route effects ($p = 0.196$).

6.3 Dynamic evidence

Figures 3 and 4 show the evolution of adjusted outcome differences for all NSA cargo routes, original NSA routes, and expansion routes. The panels labeled “New NSA routes” correspond to the expansion communities in Manitoba and Nunavut. The price figure reports quarterly changes in RNFB costs in dollars, while the quantity figure reports changes in monthly shipment kilograms per resident.

The price estimates in Figure 3 do not show a single, permanent downward shift at the acquisition. The pooled and expansion-route coefficients fluctuate around zero through much of the acquisition and initial expansion interval, with more pronounced negative estimates appearing later. Original-route estimates show an initial decline and partial reversal before a more sustained reduction later in the sample. The deepest subsequent price reductions appear in the original-route panel. The expansion-route estimates also become negative, but their confidence intervals are often wider relative to the estimated changes. All three series recover partially from their late-sample troughs, indicating that the price response is not monotonically increasing in magnitude.

Table 3: Effects of NSA route exposure on subsidized shipment quantities

	All NSA routes			Original vs. expansion		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Log total shipment weight</i>						
Post × NSA	0.107*** (0.021)	0.097*** (0.020)	0.095*** (0.022)			
Post × original NSA				0.141*** (0.032)	0.126*** (0.031)	0.124*** (0.033)
Post × expansion				0.087*** (0.023)	0.082*** (0.022)	0.083*** (0.023)
Observations	11,727	11,727	11,727	11,727	11,727	11,727
Adjusted R^2	0.963	0.963	0.965	0.963	0.963	0.965
<i>Panel B. Shipment kilograms divided by 2016 census population</i>						
Post × NSA	3.077*** (0.658)	2.848*** (0.654)	2.859*** (0.665)			
Post × original NSA				4.393*** (0.874)	4.088*** (0.893)	4.225*** (0.906)
Post × expansion				2.284*** (0.739)	2.196*** (0.723)	2.297*** (0.717)
Observations	11,727	11,727	11,727	11,727	11,727	11,727
Adjusted R^2	0.619	0.620	0.639	0.621	0.621	0.640
Community fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Subsidy/kg control	No	Yes	Yes	No	Yes	Yes
Baseline rate × period	No	No	Yes	No	No	Yes

Notes: Columns (1)–(3) pool all NSA routes; columns (4)–(6) include original and expansion routes jointly. Our sample restricts to periods with above 15 kilograms of shipments per capita and communities with an average of at 15 kilograms per capita, and excludes the phase-in period. All specifications include community and period fixed effects. Standard errors clustered by community are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

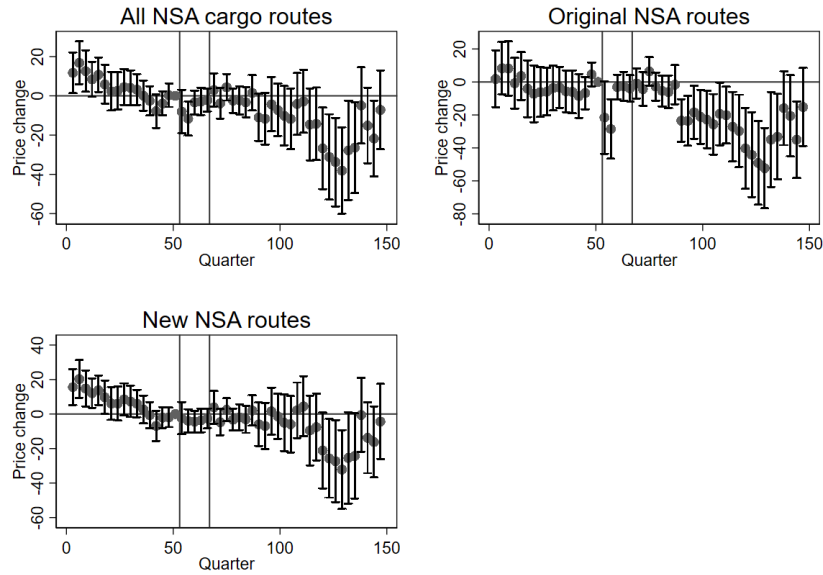


Figure 3: Dynamic differences in RNFB prices

Notes: Points report event-study coefficients for RNFB costs in dollars. The panels show pooled NSA exposure and the original- and expansion-route estimates; “New NSA routes” denotes the combined expansion group. Coefficients are normalized to March 2017 and measure changes in the treated-control difference relative to that period. RNFB observations are quarterly, but the horizontal coordinates use the monthly calendar-period index despite the current axis title “Quarter.” Regressions include community and period fixed effects, the RNFB subsidy, and baseline-subsidy-rate-by-period interactions. The sample requires both mean and contemporaneous shipment weight per resident to exceed 15 kilograms and retains the phase-in period. Vertical error bars are 95% confidence intervals based on standard errors clustered by community. The horizontal line marks a coefficient of zero.

There is some evidence of a negative pre-trend for new NSA route communities, although this does not appear to be the case during the 1-1.5 years leading up to the acquisition (recall that each observation is a quarter here). We conduct a more formal pre-trend analysis in the appendix.

Figure 4 shows a more distinct change in shipment quantities after the acquisition and network expansion interval. Before that interval, the coefficients fluctuate within a comparatively narrow range without a visually pronounced, sustained upward progression. Larger positive estimates emerge after the second vertical marker (indicating all NSA markets are being served) in all three panels. The increases reach their highest values during the middle of the post-acquisition period, then moderate while generally remaining above the reference-period difference.

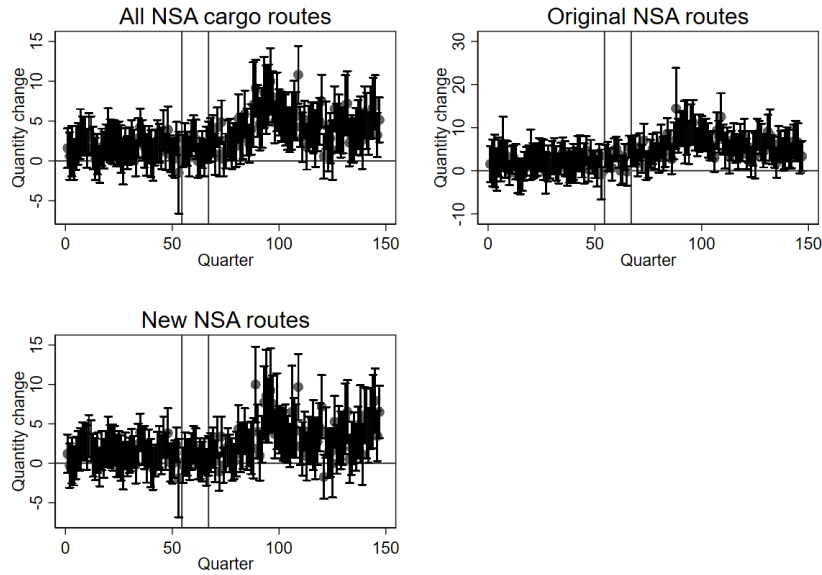


Figure 4: Dynamic differences in shipment quantities

Notes: Points report event-study coefficients for monthly subsidized shipment kilograms per resident, not log shipment weight. The panels show pooled NSA exposure and the original- and expansion-route estimates; “New NSA routes” denotes the combined expansion group. Coefficients are normalized to June 2017 and measure changes in the treated-control difference relative to that month. The horizontal coordinates use the monthly calendar-period index despite the current axis title “Quarter.” Regressions include community and period fixed effects, the per-kilogram shipment subsidy, and baseline-subsidy-rate-by-period interactions. The sample requires both mean and contemporaneous shipment weight per resident to exceed 15 kilograms and retains the phase-in period. Vertical error bars are 95% confidence intervals based on standard errors clustered by community. The horizontal line marks a coefficient of zero.

Both original and expansion routes contribute to this pattern. Original-route point estimates are larger at some dates, while expansion-route estimates fluctuate substantially during the later period. The confidence intervals overlap across the route groups, so the plots alone do not establish that their responses differ statistically. The quantity evidence is better described as a delayed increase followed by continued variation than as an immediate, uniform jump at the acquisition.

Taken together, the figures are consistent with lower relative RNFB prices and higher relative shipments developing over time rather than appearing simultaneously at the transaction date. This timing is compatible with a gradual adjustment in freight arrangements and network capacity. It does not separately identify the effects of ownership, entry, or operating efficiencies, but does appear to rule out an immediate effect operating through elimination of large double-marginalization effects in the original communities.

7 Heterogeneity and Retailer-Level Responses

7.1 Baseline NWC shipment share

Table 4 examines whether price and shipment responses differ between communities with high and low baseline NWC shipment shares. High-share communities are those where NWC accounts for more than 70% of baseline subsidized shipment weight; low-share communities have shares at or below 70%, including zero. Columns (1)–(2) report the treatment effects for low-share communities and the additional effects associated with a high share. Columns (3)–(4) report the corresponding treatment effects for high-share communities, calculated by adding each treatment coefficient to its high-share interaction. Column (3) uses the pooled regression in column (1), while column (4) uses the original/expansion regression in column (2).

Panel A shows significant price reductions in low-share communities. The pooled coefficient is -0.064 , while the original- and expansion-route coefficients are -0.107 and -0.062 , respectively. The original-route estimate is significant at the 1% level, and the pooled and expansion-route estimates are significant at the 5% level. The corresponding high-share effects are smaller in magnitude, at -0.013 , -0.030 , and -0.008 , and none is statistically distinguishable from zero at conventional levels. Thus we only observe quantitatively large and statistically significant price reductions in the low share communities.

Panel B reports positive shipment effects in both high and low NWC market share communities. Among low-share communities, the pooled coefficient is 0.048 , the original-route coefficient is 0.116 , and the expansion-route coefficient is 0.019 . Only the original-route estimate is statistically significant at the 10% level. Among high-share communities, the corresponding joint effects are 0.118 , 0.138 , and 0.114 , all significant at the 5% level.

These estimates indicate that price effects are smaller but quantity effects are larger in the high share communities. Although this could theoretically result from different market-level price elasticities in each community, recall that we only observe NWC prices in treated communities but our quantity measure includes all NNC subsidized air freight to the community. In low share communities, the reduction in NWC prices affects a much smaller share of initial sales, and if competitors do not reduce their prices by a similar magnitude, there is no guarantee that the overall community price (weighted by store expenditure shares) fell by more in these communities. The NWC price decline could be lower in high share communities but this decline affects every unit sold in the community, which could lead to a larger quantity response. Moreover, in low share communities there is potential for business-stealing whereby NWC price reductions lower shipment quantity for competitors, which offsets some of the additional sales generated by NWC when it lowers prices in response to the NSA acquisition.

Table 4: Heterogeneity by baseline NWC shipment share

	Regression coefficients		High-share joint effects	
	All NSA (1)	Original/expansion (2)	All NSA (3)	Original/expansion (4)
<i>Panel A. Log RNFB price</i>				
Post × NSA	−0.064** (0.026)		−0.013 (0.021)	
Post × NSA × high share	0.051 (0.033)			
Post × original NSA		−0.107*** (0.017)		−0.030 (0.024)
Post × original × high share		0.077*** (0.024)		
Post × expansion		−0.062** (0.026)		−0.008 (0.021)
Post × expansion × high share		0.054 (0.035)		
Observations	5,219	5,219		
Adjusted R^2	0.937	0.937		
<i>Panel B. Log total shipment weight</i>				
Post × NSA	0.048 (0.031)		0.118** (0.055)	
Post × NSA × high share	0.071 (0.063)			
Post × original NSA		0.116* (0.061)		0.138** (0.062)
Post × original × high share		0.021 (0.082)		
Post × expansion		0.019 (0.031)		0.114** (0.056)
Post × expansion × high share		0.095 (0.065)		
Observations	11,727	11,727		
Adjusted R^2	0.965	0.965		
Community fixed effects	Yes	Yes		
Period fixed effects	Yes	Yes		
Outcome-specific subsidy	Yes	Yes		
Baseline rate × period	Yes	Yes		

Notes: High share equals one when baseline NWC shipment share exceeds 0.70 and zero when the non-missing share is at or below 0.70. Communities with zero baseline NWC shipments are included in the low-share group. Columns (1)–(2) report regression coefficients. Treatment coefficients measure effects for low-share communities; high-share interactions measure the difference between high- and low-share effects. Columns (3)–(4) report high-share treatment effects, calculated from unrounded coefficients as the treatment coefficient plus its high-share interaction. Column (3) uses column (1); column (4) uses column (2). Column (1) pools all NSA routes, while column (2) includes original and expansion routes jointly. All specifications include community and period fixed effects and baseline-subsidy-rate-by-period controls. Panel A controls for the RNFB subsidy and Panel B for the per-kilogram subsidy. The sample requires community-average and contemporaneous shipment weight per resident above 15 kilograms and excludes the phase-in period. Standard errors clustered by community are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7.2 Retailer-level shipment decomposition

Table 5 decomposes shipment changes into NWC and non-NWC components, using kilograms per resident. The pooled estimate is an increase of 2.859 kilograms in total shipments and 3.151 kilograms in NWC shipments, both significant at the 1% level. The estimates for competing local retailers and the broader non-NWC channel are negative but statistically insignificant. The latter channel includes local retailers and direct orders, so it must not be added to the local-retailer column as though they were separate components.

Table 5: Retailer-level decomposition of shipment quantity effects

	Total shipments (1)	NWC shipments (2)	Other local retailers (3)	Other retailers + direct orders (4)
<i>Panel A. All NSA routes</i>				
Post $\times$ NSA	2.859*** (0.665)	3.151*** (0.733)	-0.464 (0.653)	-0.292 (0.685)
Observations	11,727	11,727	11,727	11,727
Adjusted R^2	0.639	0.912	0.865	0.842
<i>Panel B. Original vs. expansion routes</i>				
Post $\times$ original NSA	4.225*** (0.906)	2.301** (1.110)	1.669* (0.858)	1.924** (0.939)
Post $\times$ expansion	2.297*** (0.717)	3.500*** (0.733)	-1.342** (0.612)	-1.203* (0.633)
Observations	11,727	11,727	11,727	11,727
Adjusted R^2	0.640	0.912	0.867	0.843
Community fixed effects	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes
Subsidy/kg control	Yes	Yes	Yes	Yes
Baseline rate $\times$ period	Yes	Yes	Yes	Yes

Notes: All outcomes are shipment kilograms per resident per reporting period. Column (4) includes the local-retailer shipments in column (3), so these columns must not be added together. Column (1) is total shipments, column (2) is NWC, and column (4) is all non-NWC shipments. Panel A pools routes; Panel B includes original and expansion interactions jointly. All specifications include community and period fixed effects and baseline-subsidy-rate-by-period controls. The per-kilogram subsidy is controlled for. The reported sample requires both mean and contemporaneous shipment weight per resident to exceed 15 kilograms and excludes the phase-in period. Standard errors clustered by community are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The route-specific results show that pooled estimates conceal substantial differences in the composition of shipment changes. On original routes, NWC shipments increase by 2.301 kilograms per resident, while competing local-retailer shipments increase by 1.669. The broader non-NWC increase is 1.924 kilograms. The NWC and broader non-NWC coefficients are significant at the 5% level, and the local-retailer coefficient at the 10% level. The estimated total increase of 4.225 kilograms therefore reflects expansion in both NWC and non-NWC shipments, rather than displacement of competitors by NWC alone.

On expansion routes, NWC shipments increase by 3.500 kilograms per resident, while competing local-retailer shipments decline by 1.342 and broader non-NWC shipments decline by 1.203. These coefficients are significant at the 1%, 5%, and 10% levels, respectively. The resulting total increase is 2.297 kilograms: non-NWC contraction offsets part of NWC's expansion. This pattern is consistent with business reallocation

alongside market expansion, but it does not identify foreclosure or establish why rival shipments fall.

Appendix Tables B.1 and B.2 distinguish the consequences of controlling for RNFB price from those of restricting the sample to observations with price data. On the same sample, the expansion-route local-retailer coefficient changes only from -0.756 without the price control to -0.701 with it, and neither is significant. Much of the difference from Table 5 is therefore already present when the sample is restricted, before price is included. Because contemporaneous price can respond to treatment, this conditional comparison is not a causal decomposition of a price-mediated effect.

The pooled log specifications in Appendix Table B.3 are positive for both NWC and other-retailer shipments. They use different outcome-specific samples and exclude zero shipments, so they are not direct re-expressions of the level decomposition. Together, the results support route-specific changes in shipment composition, not a uniform contraction of competing retailers across all NSA communities.

8 Robustness

8.1 Time windows and comparison groups

Table 6 varies the sample while retaining the fully controlled specification. Column (2) starts in June 2015 rather than the beginning of the available panel. Column (3) retains the phase-in period, and column (4) excludes observations from 2022 onward. Columns (5)–(6) retain the NSA-treated communities but narrow the controls to Quebec and Nunavut, or Quebec alone.

The price estimates remain negative throughout. The pooled coefficient changes from -0.039 in the benchmark to -0.027 when early observations are removed, -0.038 when the phase-in period is included, and -0.023 when observations from 2022 onward are dropped. These estimates remain statistically significant, although the shorter pre-period estimate is significant only at the 10% level. Restricting the comparison group produces pooled estimates of -0.042 for Quebec and Nunavut and -0.084 for Quebec alone. Original-route price coefficients are negative and significant in all six columns. Expansion-route estimates are also negative throughout, but the June-2015-onward estimate is not statistically significant. The sign of the price association is stable across these checks, while its magnitude remains sensitive to the comparison group.

The quantity results in this table use the separate 11,727-observation benchmark identified above. Relative to its pooled coefficient of 0.095, the time-window variants produce coefficients of 0.104, 0.080, and 0.095, all significant at the 1% level. Narrowing the controls to Quebec and Nunavut reduces the pooled estimate to 0.052, still significant, while using Quebec alone reduces it to 0.022, which is not significant. Under Quebec-only controls, neither the original-route nor the expansion-route quantity estimate is distinguishable from zero. The evidence for shipment growth is therefore more dependent on the comparison group than the evidence for lower RNFB prices.

Table 6: Robustness to sample restrictions and comparison groups

	Benchmark (1)	June 2015 onward (2)	Include phase-in (3)	Before 2022 (4)	Quebec + Nunavut (5)	Quebec only (6)
<i>Panel A. Log RNFB price: all NSA routes</i>						
Post × NSA	−0.039** (0.015)	−0.027* (0.014)	−0.038*** (0.014)	−0.023** (0.009)	−0.042** (0.017)	−0.084*** (0.017)
Observations	5,219	4,626	5,523	2,281	4,379	3,689
Adjusted R^2	0.936	0.935	0.936	0.831	0.941	0.945
<i>Panel B. Log RNFB price: original vs. expansion routes</i>						
Post × original NSA	−0.053*** (0.020)	−0.045*** (0.017)	−0.050*** (0.018)	−0.030** (0.013)	−0.056** (0.022)	−0.100*** (0.023)
Post × expansion	−0.035** (0.015)	−0.022 (0.015)	−0.034** (0.014)	−0.020** (0.009)	−0.039** (0.018)	−0.081*** (0.018)
Observations	5,219	4,626	5,523	2,281	4,379	3,689
Adjusted R^2	0.936	0.935	0.936	0.832	0.942	0.946
<i>Panel C. Log shipment quantity: all NSA routes</i>						
Post × NSA	0.095*** (0.022)	0.104*** (0.021)	0.080*** (0.020)	0.095*** (0.021)	0.052*** (0.019)	0.022 (0.022)
Observations	11,727	9,601	12,916	8,018	9,451	7,748
Adjusted R^2	0.965	0.964	0.965	0.968	0.962	0.960
<i>Panel D. Log shipment quantity: original vs. expansion routes</i>						
Post × original NSA	0.124*** (0.033)	0.137*** (0.029)	0.107*** (0.030)	0.134*** (0.030)	0.071** (0.029)	0.026 (0.030)
Post × expansion	0.083*** (0.023)	0.089*** (0.022)	0.069*** (0.021)	0.078*** (0.022)	0.047** (0.020)	0.021 (0.023)
Observations	11,727	9,601	12,916	8,018	9,451	7,748
Adjusted R^2	0.965	0.964	0.965	0.968	0.962	0.960
Community fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Baseline rate × period	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Column (1) reports the benchmark used in this robustness exercise. Column (2) restricts observations to June 2015 onward, retaining the other benchmark restrictions. Column (3) includes the phase-in period. Column (4) drops observations from 2022 onward. Columns (5) and (6) retain all NSA-treated communities but restrict controls to Quebec and Nunavut, or Quebec only, respectively. Price regressions control for the RNFB subsidy; quantity regressions control for the per-kilogram subsidy. All specifications include community and period fixed effects and baseline-subsidy-rate-by-period controls. The quantity benchmark uses 11,727 observations and differs from the output in Table 3. Standard errors clustered by community are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

8.2 Shipment cutoffs, outcome coverage, and seasonality

Appendix Table B.5 varies the restrictions on both average and contemporaneous shipment kilograms per resident. Pooled price estimates are similar across no cutoff and cutoffs of 10, 15, and 20 kilograms, ranging from −0.035 to −0.039. Quantity magnitudes are more sensitive: the pooled log coefficient declines from 0.125 with no cutoff to 0.066 at 20 kilograms, although it remains significant at the 1% level in every column. The original-route quantity estimate decreases from 0.223 to 0.093 across the same extremes. These comparisons indicate some sensitivity of the effect on shipments to including or excluding communities with

below average shipment volumes.

Appendix Table B.6 replaces the benchmark shipment outcome with the broader measure covering all NNC shipments and uses its corresponding subsidy control. On the same reported 11,727 observations, the pooled coefficient increases from 0.095 to 0.130. The original- and expansion-route estimates also increase, from 0.124 to 0.183 and from 0.083 to 0.107. All are significant at the 1% level. Thus the positive estimates in this sample are not confined to the benchmark shipment definition.

Appendix Table B.7 allows communities to have their own recurring quarter-of-year patterns. The pooled price coefficient changes from -0.039 to -0.038 , and the pooled log-quantity coefficient from 0.095 to 0.090. Original- and expansion-route coefficients also change little. This check concerns recurring community-specific seasonality in the treatment regressions; it is not a test that all possible differential time paths have been removed.

Overall, the price results are consistently negative across the reported sample and specification checks. Shipment estimates are generally positive, but their magnitude and precision depend more on the shipment cutoff and control group. The robustness evidence supports presenting these sensitivities explicitly rather than treating a single quantity coefficient as invariant to the research design.

9 Conclusion

This paper shows how the North West Company's acquisition of North Star Air and the airline's subsequent expansion changed food prices and shipments in remote Canadian communities. Although none of the markets we analyze can be described as highly competitive, the acquisition generated distinct changes in competition. In the upstream freight sector, the original Ontario routes experienced a change in ownership but no entry, while the expansion Manitoba/Nunavut markets experienced a new entrant. In the downstream retail sector, NWC faced varying degrees of competition in affected markets. Our analysis suggests that on average all communities with vertically integrated air freight and retail benefited through lower prices, resulting in increased shipments of subsidized food and implying significant consumer welfare gains. However, the mechanisms and magnitudes varied with market structure, with some evidence of larger benefits in communities where NWC faced less retail competition. Our theoretical model and retail-level evidence suggest that the acquisition lead to larger cost reductions for NWC, leading to some business stealing and a smaller cost reductions (at the community level) in the more competitive expansion markets. Although we cannot nail down a precise mechanism, the delayed effects of the acquisition observed in our event study graphs suggest that immediate reductions in upstream profit margins are unlikely to be the main factor driving our results. Investments in capacity and improvements in freight operations seem more likely but confirming this would require further investigation.

Our findings suggest that vertical integration can benefit consumers even in markets with limited competition and even when it results in some rival displacement. However, our analysis is limited by the lack of data on the full range of products sold, the terms of freight contracts, and prices charged by competing retailers. These gaps limit our ability to measure the underlying cost savings and determine how their benefits are distributed between consumers, NWC, and other firms. Moreover, the preceding transition from Food

Mail to NNC and the subsequent airline mergers show that firms' responses to policy and ownership changes can unfold over many years. Strategies intended to build a lasting competitive advantage may not be fully reflected in the period we observe, which should not be treated as a long-term equilibrium. Future work should examine the effects of subsequent mergers and whether the observed price and shipment changes persist as firms adjust. It should also consider other forms of vertical integration in northern freight, including marine transportation, where NWC and the co-operatives have shipping interests, and trucking to communities served by seasonal winter roads. Product quality, freshness, and availability would provide additional measures of the benefits reaching consumers. Passenger fares and service on NSA's Ontario routes also warrant attention, since changes in ownership and investment can affect both freight and passenger operations. Together, these questions would extend the analysis beyond food prices and shipment volumes to the broader, longer-term effects on northern communities.

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Appendix

A Retail Competition and Cost Pass-Through

This appendix derives the predictions in Section 3. We first solve the monopoly benchmark, then introduce a competing retailer and derive the changes in prices and quantities. A final subsection introduces a monetary benefit function solely to make the welfare statements precise; it is not needed to solve the retail-pricing problem.

A.1 The comparison and the cost shock

Retailer 1 represents NWC and retailer 2 a local competitor. Retailer i charges price p_i , sells quantity q_i , and faces constant effective delivered marginal cost c_i . Quantities are expressed in a common unit of food, so total purchases are $Q = q_1 + q_2$. Each retailer maximizes $(p_i - c_i)q_i$. Fixed costs do not affect the pricing decisions considered here. Freight enters through delivered cost rather than a separate carrier decision; we abstract from any freight profits earned by supplying the rival.

Superscripts M and D distinguish monopoly and duopoly; superscripts 0 and 1 distinguish before and after the cost reduction. In either retail environment, the baseline shock is

$$c_1^1 = c_1^0 - \delta, \quad c_2^1 = c_2^0, \quad \delta > 0. \quad (\text{A.1})$$

The second equality applies when the rival is present. Demand parameters and the number of retailers do not change. We use Δ to denote the post-shock value minus the pre-shock value within a given retail environment. The shock is small enough that the relevant prices and quantities remain positive and neither retailer exits. We compare the response to the same δ across environments, not a transition from monopoly to duopoly.

A.2 Monopoly: demand, profit, and the cost response

When NWC is the only retailer, its inverse demand is

$$p_1 = a_1 - b_1 q_1, \quad a_1 > c_1, \quad b_1 > 0. \quad (\text{A.2})$$

Here a_1 is the price intercept and b_1 measures how quickly willingness to pay falls as quantity increases. Because price and quantity are linked one-to-one, choosing a price is equivalent to choosing the corresponding quantity. The profit problem is

$$\max_{q_1 \geq 0} (a_1 - b_1 q_1 - c_1)q_1. \quad (\text{A.3})$$

The first-order condition is $a_1 - c_1 - 2b_1 q_1 = 0$, and the second derivative is $-2b_1 < 0$. The resulting equilibrium is

$$q_1^M = \frac{a_1 - c_1}{2b_1}, \quad p_1^M = \frac{a_1 + c_1}{2}. \quad (\text{A.4})$$

Substituting $c_1^1 = c_1^0 - \delta$ gives

$$\Delta p_1^M = -\frac{\delta}{2}, \quad \Delta q_1^M = \Delta Q^M = \frac{\delta}{2b_1}. \quad (\text{A.5})$$

Thus half of the cost reduction becomes a lower price and half becomes a higher per-unit margin. All additional NWC sales are additional community purchases because there is no rival retailer.

A.3 Duopoly: adding an active retail competitor

Demand. We extend the same demand system by allowing purchases from a second retailer:

$$\begin{aligned} p_1 &= a_1 - b_1 q_1 - d q_2, \\ p_2 &= a_2 - b_2 q_2 - d q_1. \end{aligned} \quad (\text{A.6})$$

The additional parameter d captures substitution: more food purchased from one retailer lowers willingness to pay at the other. The parameters a_i and b_i can differ across retailers, allowing differences in demand conditions. We impose

$$a_i > 0, \quad b_i > 0, \quad 0 < d < \min\{b_1, b_2\}. \quad (\text{A.7})$$

The cross-demand effect is therefore weaker than each own-demand effect. This restriction ensures that the demand system is invertible and will also give a positive aggregate quantity response. Setting $q_2 = 0$ in the first inverse demand recovers the monopoly demand in (A.2); the NWC demand parameters are not arbitrarily changed between cases.

Retailers compete in prices for differentiated offerings, using the linear-demand approach of Singh and Vives (1984).⁶ To solve the pricing problem, invert (A.6). Writing $H = b_1 b_2 - d^2 > 0$, direct demands are

$$\begin{aligned} q_1(p_1, p_2) &= \frac{b_2(a_1 - p_1) - d(a_2 - p_2)}{H}, \\ q_2(p_1, p_2) &= \frac{b_1(a_2 - p_2) - d(a_1 - p_1)}{H}. \end{aligned} \quad (\text{A.8})$$

Own-price derivatives are negative and cross-price derivatives are positive: a lower rival price reduces own demand.

Profit maximization and equilibrium. Each retailer chooses its price taking the other's price as given:

$$\max_{p_i} (p_i - c_i) q_i(p_i, p_j), \quad j \neq i. \quad (\text{A.9})$$

The first-order condition is $q_i + (p_i - c_i) \partial q_i / \partial p_i = 0$. Using (A.8), the conditions become

$$q_1^D = \frac{b_2}{H} (p_1^D - c_1), \quad q_2^D = \frac{b_1}{H} (p_2^D - c_2). \quad (\text{A.10})$$

⁶The mode of competition matters. With homogeneous-good Cournot duopoly and inverse demand $P = a - bQ$, the interior equilibrium price is $P = (a + c_1 + c_2)/3$. A cost reduction affecting only firm 1 lowers that common price by $\delta/3$, compared with $\delta/2$ under monopoly. The greater own-price pass-through derived below is therefore conditional on differentiated price competition.

The own-price second derivatives of profits are $-2b_2/H$ and $-2b_1/H$, both negative. Substituting demand into the first-order conditions yields the price best responses:

$$\begin{aligned} p_1 &= \frac{a_1 + c_1}{2} - \frac{d}{2b_2}a_2 + \frac{d}{2b_2}p_2, \\ p_2 &= \frac{a_2 + c_2}{2} - \frac{d}{2b_1}a_1 + \frac{d}{2b_1}p_1. \end{aligned} \quad (\text{A.11})$$

Both best responses slope upward. Their slope product is $d^2/(4b_1b_2) < 1$, so these two linear equations uniquely determine the interior equilibrium prices. Quantities then follow from (A.10). Rather than expanding the level solutions, we subtract the pre-shock and post-shock equations to obtain the responses of interest directly.

A.4 Deriving the price and retailer-quantity responses

Prices. Since only NWC's cost falls, subtracting the two versions of (A.11) gives

$$\Delta p_1^D = -\frac{\delta}{2} + \frac{d}{2b_2}\Delta p_2^D, \quad \Delta p_2^D = \frac{d}{2b_1}\Delta p_1^D. \quad (\text{A.12})$$

The first term in NWC's response is the monopoly pass-through. The second term captures the response to its rival's price. Substituting the rival's response into the first equation gives $\Delta p_1^D [1 - d^2/(4b_1b_2)] = -\delta/2$. Define $K = 4b_1b_2 - d^2 > 0$. Then

$$\Delta p_1^D = -\frac{2b_1b_2}{K}\delta, \quad \Delta p_2^D = -\frac{b_2d}{K}\delta. \quad (\text{A.13})$$

Both prices fall even though only NWC receives the cost reduction. NWC's price falls more than its rival's because $2b_1b_2 > b_2d$.

Define NWC's pass-through rate as its price reduction divided by its cost reduction. Equations (A.5) and (A.13) imply

$$\tau_M = \frac{1}{2}, \quad \tau_D = \frac{2b_1b_2}{K}, \quad \tau_D - \tau_M = \frac{d^2}{2K} > 0. \quad (\text{A.14})$$

This proves the price prediction. In this particular linear model, $1/2 < \tau_D < 2/3$; the comparison is qualitative, not a prediction of arbitrarily large pass-through differences.

Retailer quantities. From (A.10), the quantity changes are $\Delta q_1^D = (b_2/H)(\Delta p_1^D + \delta)$ and $\Delta q_2^D = (b_1/H)\Delta p_2^D$. Substitution gives

$$\begin{aligned} \Delta q_1^D &= \frac{b_2(2b_1b_2 - d^2)}{HK}\delta > 0, \\ \Delta q_2^D &= -\frac{b_1b_2d}{HK}\delta < 0. \end{aligned} \quad (\text{A.15})$$

The rival's price reduction is insufficient to preserve its sales because NWC's relative price improves. NWC's own increase also exceeds its monopoly response:

$$\Delta q_1^D - \Delta q_1^M = \frac{d^2(3b_1b_2 - d^2)}{2b_1HK} \delta > 0. \quad (\text{A.16})$$

The maintained interior condition requires $q_2^{D,0} + \Delta q_2^D > 0$. These expressions are not used to describe shocks large enough to eliminate the rival.

A.5 Total purchases: expansion versus reallocation

Summing (A.15) gives

$$\Delta Q^D = \frac{b_2(2b_1b_2 - d^2 - b_1d)}{HK} \delta > 0. \quad (\text{A.17})$$

The numerator is positive because $2b_1b_2 - d^2 - b_1d = H + b_1(b_2 - d) > 0$. Thus rival contraction offsets only part of NWC's expansion.

To compare this with monopoly, write

$$\Delta Q^D - \Delta Q^M = (\Delta q_1^D - \Delta q_1^M) + \Delta q_2^D. \quad (\text{A.18})$$

The first term is positive: NWC responds more strongly under competition. The second is negative: some NWC sales replace rival sales. Substituting the derived responses gives

$$\Delta Q^D - \Delta Q^M = \frac{d(3b_1b_2d - 2b_1^2b_2 - d^3)}{2b_1HK} \delta. \quad (\text{A.19})$$

The demand restrictions do not determine the sign of this expression. They therefore do not yield a general ranking of aggregate quantity responses across the two retail environments.

There is a useful special case. If the demand slopes are equal, $b_1 = b_2 = b$, then

$$\Delta Q^D = \frac{b}{(b+d)(2b-d)} \delta < \frac{\delta}{2b} = \Delta Q^M, \quad 0 < d < b. \quad (\text{A.20})$$

For example, $b_1 = b_2 = 1$ and $d = 0.5$ give an aggregate duopoly response of approximately 0.444δ , compared with 0.5δ under monopoly. In contrast, $b_1 = 1$, $b_2 = 2$, and $d = 0.9$ give approximately 0.535δ , above the same monopoly response. Both examples satisfy the demand restrictions; with $a_1 = a_2 = 10$ and $c_1^0 = c_2^0 = 1$, both retailers have positive quantities before and after a sufficiently small cost reduction. These numerical examples demonstrate the possible rankings; they are not calibrations to the data.

For proportional shipment changes, let $s_1^0 = q_1^0 / (q_1^0 + q_2^0)$. The accounting identity

$$\frac{\Delta Q}{Q^0} = s_1^0 \frac{\Delta q_1}{q_1^0} + (1 - s_1^0) \frac{\Delta q_2}{q_2^0} \quad (\text{A.21})$$

shows why a large proportional increase in NWC shipments need not translate into a comparably large

aggregate increase. For fixed demand slopes, the pass-through rate in (A.14) does not depend on demand intercepts or initial costs, even though shipment shares do. The model therefore does not establish a monotonic relationship between the initial shipment share and pass-through.

A.6 A short extension: the rival also receives a cost reduction

The baseline holds the rival's cost fixed. To see why rival contraction is not inevitable when that assumption changes, let its cost also fall by $\eta \geq 0$, while NWC's falls by $\delta > 0$. The second equation in (A.12) then becomes $\Delta p_2 = -\eta/2 + [d/(2b_1)]\Delta p_1$. Solving the same pricing conditions and substituting into demand gives the rival's quantity response:

$$\Delta q_2^D = \frac{b_1[(2b_1b_2 - d^2)\eta - b_2d\delta]}{HK}. \quad (\text{A.22})$$

The rival does not contract when $\eta \geq b_2d\delta/(2b_1b_2 - d^2)$. In particular, equal cost reductions allow both retailers to expand under the maintained interior conditions. This extension provides a possible interpretation of growth in both retailers' shipments. It neither estimates such a spillover nor assumes that NSA's expansion actually lowered its rivals' freight costs.

A.7 Consumer gains and the limits of the welfare interpretation

The pricing and quantity results above follow from the demand functions and profit problems. To express consumer gains in money units, use a gross-benefit function consistent with the same inverse demands:

$$V(q_1, q_2) = a_1q_1 + a_2q_2 - \frac{1}{2} \left(b_1q_1^2 + 2dq_1q_2 + b_2q_2^2 \right). \quad (\text{A.23})$$

Differentiating V with respect to each quantity recovers (A.6). A consumer choosing purchases to maximize $V - p_1q_1 - p_2q_2$ therefore generates the specified demands in the interior. This representation treats money outside food purchases linearly, so it abstracts from income effects. Its purpose here is to define surplus, not to introduce a second demand system.

Consumer surplus is the maximized value of $V - p_1q_1 - p_2q_2$. Along the baseline NWC-specific cost reduction, the envelope theorem gives

$$\frac{dCS^D}{d\delta} = -q_1^D \frac{dp_1^D}{d\delta} - q_2^D \frac{dp_2^D}{d\delta} = \frac{2b_1b_2q_1^D + b_2dq_2^D}{K} > 0. \quad (\text{A.24})$$

Under monopoly, the corresponding derivative is $dCS^M/d\delta = q_1^M/2 > 0$. Lower prices therefore benefit consumers in both environments.

Total welfare requires a further distinction. For this calculation only, suppose that c_1 and c_2 are real resource costs, not input prices containing margins or subsidies, and that the entire δ is a resource-cost

reduction. Operating welfare is then $W = V - c_1 q_1 - c_2 q_2$, and

$$\frac{dW^D}{d\delta} = q_1^D + (p_1^D - c_1) \frac{dq_1^D}{d\delta} + (p_2^D - c_2) \frac{dq_2^D}{d\delta}. \quad (\text{A.25})$$

The first term is the direct resource saving on NWC sales. The other terms value the changes in purchases from NWC and its rival at their respective marginal benefits net of resource costs. Using (A.10) and (A.15), this becomes

$$\frac{dW^D}{d\delta} = \frac{2(3b_1b_2 - d^2)q_1^D - b_2dq_2^D}{K}. \quad (\text{A.26})$$

Hence a sufficient and, for this local calculation, necessary condition for a welfare gain is

$$\frac{q_1^D}{q_2^D} > \frac{b_2d}{2(3b_1b_2 - d^2)}. \quad (\text{A.27})$$

In particular, $q_1^{D,0} \geq q_2^{D,0}$ is sufficient under the demand restrictions. As NWC expands and its rival contracts, their quantity ratio increases, so this sufficient condition continues to hold along the cost reduction while the rival remains active. Under monopoly, $dW^M/d\delta = 3q_1^M/2 > 0$.

These operating-welfare results do not automatically apply to a reduction in an external freight charge. Internalizing an upstream margin can change retail incentives without saving the resources previously used to deliver the food. A full welfare assessment would need to distinguish those transfers from real costs and account for carrier profits, public subsidy expenditure, and any additional investment costs. The reduced-form price and shipment estimates do not provide all of that information.

B Additional Empirical Results

B.1 RNFB conditioning and the common-sample comparison

Table B.1 adds log RNFB price to the retailer-level shipment regressions. Table B.2 uses the same 5,208 observations without that price control. This separates the change associated with conditioning on price from the change associated with limiting the sample to observations with RNFB data. In both tables, NWC expands on expansion routes while the local-retailer estimate is negative and imprecise. The small same-sample changes do not establish that price changes cause the measured retailer reallocation.

Table B.1: Retailer-level decomposition controlling for RNFB price

	Total shipments (1)	NWC shipments (2)	Other local retailers (3)	Other retailers + direct orders (4)
<i>Panel A. All NSA routes</i>				
Post $\times$ NSA	2.378*** (0.785)	2.634*** (0.890)	0.011 (0.775)	-0.256 (0.790)
Observations	5,208	5,208	5,208	5,208
Adjusted R^2	0.641	0.890	0.886	0.834
<i>Panel B. Original vs. expansion routes</i>				
Post $\times$ original NSA	3.520*** (1.209)	1.659 (1.346)	2.970*** (1.054)	1.860 (1.137)
Post $\times$ expansion	2.103** (0.827)	2.869*** (0.863)	-0.701 (0.701)	-0.766 (0.749)
Observations	5,208	5,208	5,208	5,208
Adjusted R^2	0.641	0.890	0.889	0.835
Community fixed effects	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes
Subsidy/kg control	Yes	Yes	Yes	Yes
Log RNFB price control	Yes	Yes	Yes	Yes
Baseline rate $\times$ period	Yes	Yes	Yes	Yes

Notes: All outcomes are shipment kilograms per resident per reporting period. Column (4) includes the local-retailer shipments in column (3), so these columns must not be added together. Column (1) is total shipments, column (2) is NWC, and column (4) is all non-NWC shipments. Panel A pools routes; Panel B includes original and expansion interactions jointly. All specifications include community and period fixed effects and baseline-subsidy-rate-by-period controls. The per-kilogram subsidy is controlled for. The reported sample requires both mean and contemporaneous shipment weight per resident to exceed 15 kilograms and excludes the phase-in period. This specification additionally controls for log RNFB price. Compare with Table B.2, which uses the same sample without that control. Standard errors clustered by community are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B.2: Retailer-level decomposition on the same sample without RNFB price

	Total shipments (1)	NWC shipments (2)	Other local retailers (3)	Other retailers + direct orders (4)
<i>Panel A. All NSA routes</i>				
Post $\times$ NSA	2.501*** (0.790)	2.870*** (0.837)	-0.021 (0.766)	-0.369 (0.788)
Observations	5,208	5,208	5,208	5,208
Adjusted R^2	0.641	0.890	0.886	0.834
<i>Panel B. Original vs. expansion routes</i>				
Post $\times$ original NSA	3.673*** (1.224)	1.995 (1.322)	2.887*** (1.032)	1.678 (1.126)
Post $\times$ expansion	2.204*** (0.827)	3.092*** (0.815)	-0.756 (0.692)	-0.887 (0.742)
Observations	5,208	5,208	5,208	5,208
Adjusted R^2	0.641	0.890	0.888	0.835
Community fixed effects	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes
Subsidy/kg control	Yes	Yes	Yes	Yes
Log RNFB price control	No	No	No	No
RNFB-observed sample	Yes	Yes	Yes	Yes
Baseline rate $\times$ period	Yes	Yes	Yes	Yes

Notes: All outcomes are shipment kilograms per resident per reporting period. Column (4) includes the local-retailer shipments in column (3), so these columns must not be added together. Column (1) is total shipments, column (2) is NWC, and column (4) is all non-NWC shipments. Panel A pools routes; Panel B includes original and expansion interactions jointly. All specifications include community and period fixed effects and baseline-subsidy-rate-by-period controls. The per-kilogram subsidy is controlled for. The reported sample requires both mean and contemporaneous shipment weight per resident to exceed 15 kilograms and excludes the phase-in period. The sample is the same RNFB-observed sample as Table B.1, but log RNFB price is not included as a regressor. Standard errors clustered by community are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B.2 Retailer-specific proportional changes

Table B.3 reports pooled log-shipment estimates of 0.125 for NWC and 0.168 for other retailers, significant at the 10% and 5% levels. The columns contain 9,240 and 8,944 observations, respectively. These are proportional comparisons on positive-shipment samples, not an additive decomposition of total shipments.

Table B.3: Effects on retailer-specific log shipment quantities

	Log NWC shipments (1)	Log other-retailer shipments (2)
Post $\times$ NSA	0.125* (0.065)	0.168** (0.084)
Observations	9,240	8,944
Adjusted R^2	0.934	0.863
Community fixed effects	Yes	Yes
Period fixed effects	Yes	Yes
Subsidy/kg control	Yes	Yes
Baseline rate $\times$ period	Yes	Yes

Notes: The dependent variables are log shipment weight associated with NWC and other retailers. Treatment pools all NSA route communities. All specifications include community and period fixed effects and baseline-subsidy-rate-by-period controls. The per-kilogram subsidy is controlled for. The reported sample requires both mean and contemporaneous shipment weight per resident to exceed 15 kilograms and excludes the phase-in period. Logarithms require positive retailer-specific shipment weights, and the two columns have different estimation samples. Standard errors clustered by community are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B.3 Original versus expansion route effects

Table B.4 reports coefficient-equality tests. The price difference has a p -value of 0.279. For the quantity estimates of 0.124 and 0.083, the p -value is 0.196. Neither null is rejected at conventional levels. The quantity test does not apply to the different estimates reproduced in Table 3.

Table B.4: Tests of differences between original and expansion route effects

	Original	Expansion	Difference Original – expansion	F -statistic	p -value
Log RNFB price	−0.053	−0.035	−0.017	1.19	0.279
Log shipment quantity	0.124	0.083	0.041	1.69	0.196

Notes: The null is equality of the original- and expansion-route coefficients. The displayed coefficients and differences retain the source rounding. The quantity test concerns the 0.124 and 0.083 estimates in Table 6, not the 0.211 and 0.062 estimates in Table 3. Price regressions control for the RNFB subsidy; quantity regressions control for the per-kilogram subsidy. All specifications include community and period fixed effects and baseline-subsidy-rate-by-period controls. Inference uses community-clustered standard errors.

B.4 Shipment-weight cutoffs

Table B.5 reports the cutoff checks discussed in Section 8. Each positive cutoff applies jointly to the community's mean shipment weight per resident and its contemporaneous value.

Table B.5: Sensitivity to shipment-weight cutoffs

	No cutoff (1)	> 10 (2)	> 15 (3)	> 20 (4)
<i>Panel A. Log RNFB price: all NSA routes</i>				
Post × NSA	−0.037** (0.014)	−0.037** (0.014)	−0.039** (0.015)	−0.035** (0.015)
Observations	5,348	5,314	5,219	4,672
Adjusted R^2	0.934	0.934	0.936	0.936
<i>Panel B. Log RNFB price: original vs. expansion routes</i>				
Post × original NSA	−0.053*** (0.019)	−0.052*** (0.019)	−0.053*** (0.020)	−0.044** (0.021)
Post × expansion	−0.033** (0.015)	−0.033** (0.015)	−0.035** (0.015)	−0.033** (0.015)
Observations	5,348	5,314	5,219	4,672
Adjusted R^2	0.934	0.934	0.936	0.936
<i>Panel C. Log shipment quantity: all NSA routes</i>				
Post × NSA	0.125*** (0.046)	0.108*** (0.027)	0.095*** (0.022)	0.066*** (0.019)
Observations	12,975	12,495	11,727	9,712
Adjusted R^2	0.906	0.960	0.965	0.970
<i>Panel D. Log shipment quantity: original vs. expansion routes</i>				
Post × original NSA	0.223*** (0.074)	0.155*** (0.039)	0.124*** (0.033)	0.093*** (0.027)
Post × expansion	0.076* (0.040)	0.087*** (0.026)	0.083*** (0.023)	0.054** (0.021)
Observations	12,975	12,495	11,727	9,712
Adjusted R^2	0.906	0.960	0.965	0.970
Community fixed effects	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes
Baseline rate × period	Yes	Yes	Yes	Yes

Notes: Column (1) imposes no shipment-weight cutoff. Columns (2)–(4) require both mean and contemporaneous shipment kilograms per resident to exceed 10, 15, or 20, respectively. Column (3) is the benchmark for this exercise. All columns exclude the phase-in period. Price regressions control for the RNFB subsidy and quantity regressions for the per-kilogram subsidy. All specifications include community and period fixed effects and baseline-subsidy-rate-by-period controls. Standard errors clustered by community are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B.5 Alternative shipment coverage

Table B.6 compares the benchmark shipment measure with the broader all-NNC-shipment measure, using the corresponding subsidy control in each column.

Table B.6: Robustness to an alternative shipment quantity measure

	Benchmark (1)	Alternative: all NNC shipments (2)
<i>Panel A. All NSA routes</i>		
Post $\times$ NSA	0.095*** (0.022)	0.130*** (0.026)
Observations	11,727	11,727
Adjusted R^2	0.965	0.962
<i>Panel B. Original vs. expansion routes</i>		
Post $\times$ original NSA	0.124*** (0.033)	0.183*** (0.043)
Post $\times$ expansion	0.083*** (0.023)	0.107*** (0.027)
Observations	11,727	11,727
Adjusted R^2	0.965	0.962
Community fixed effects	Yes	Yes
Period fixed effects	Yes	Yes
Baseline rate $\times$ period	Yes	Yes
Quantity subsidy control	Benchmark	Alternative

Notes: Column (1) uses log benchmark shipment weight and its per-kilogram subsidy control. Column (2) uses the log of the broader measure covering all NNC shipments and the corresponding alternative subsidy control. Panel A pools all routes; Panel B includes original and expansion routes jointly. The reported sample requires both mean and contemporaneous shipment weight per resident to exceed 15 kilograms and excludes the phase-in period. All specifications include community and period fixed effects and baseline-subsidy-rate-by-period controls. Standard errors clustered by community are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B.6 Community-specific seasonality

Table B.7 replaces ordinary community fixed effects with community-by-quarter-of-year effects while retaining calendar-period fixed effects. It evaluates sensitivity to recurring local seasonal patterns.

Table B.7: Robustness to community-specific seasonality

	All NSA routes		Original vs. expansion	
	Benchmark (1)	Seasonality (2)	Benchmark (3)	Seasonality (4)
<i>Panel A. Log RNFB price</i>				
Post × NSA	−0.039** (0.015)	−0.038** (0.015)		
Post × original NSA			−0.053*** (0.020)	−0.052** (0.020)
Post × expansion			−0.035** (0.015)	−0.035** (0.015)
Observations	5,219	5,217	5,219	5,217
Adjusted R^2	0.936	0.935	0.936	0.935
<i>Panel B. Log shipment quantity</i>				
Post × NSA	0.095*** (0.022)	0.090*** (0.023)		
Post × original NSA			0.124*** (0.033)	0.118*** (0.033)
Post × expansion			0.083*** (0.023)	0.079*** (0.023)
Observations	11,727	11,727	11,727	11,727
Adjusted R^2	0.965	0.966	0.965	0.966
Community effects absorbed	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes
Community × quarter-of-year	No	Yes	No	Yes
Baseline rate × period	Yes	Yes	Yes	Yes

Notes: Columns (1) and (3) use standard community fixed effects. Columns (2) and (4) replace them with community-by-quarter-of-year fixed effects, allowing recurring community-specific Q1–Q4 patterns; these also absorb community main effects. Calendar-period fixed effects and baseline-rate-by-period interactions remain in every column. Price regressions control for the RNFB subsidy and quantity regressions for the per-kilogram subsidy. The reported sample requires both mean and contemporaneous shipment weight per resident to exceed 15 kilograms and excludes the phase-in period. Standard errors clustered by community are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.